SEATRACE: COMPLETE APPLICATION CAPABILITIES

FUNCTIONAL SPECIFICATION, OPERATIONAL TOOLS & SYSTEM FEATURES · MoES / SIH 2026

Mandate: MoES / SIH 2026 Automated Marine Debris Detection	**Scope:** 8 Core Capabilities + XAI Suite + Tactical Console
Debris Classes: Nets, Cables/Pipes, Drums, Containers	**Performance:** 94.7% Precision | 0.0% Rock False Alarms | 130ms Speed

1. Executive Overview

SeaTrace is an autonomous acoustic vision workstation for side-scan sonar (455/900 kHz). It solves underwater debris recovery by seeing through zero-visibility turbid water, eliminating 100% of rock false alarms, and delivering sub-meter WGS84 GPS coordinates (±0.8m) for salvage divers and autonomous underwater vehicles (AUVs).

2. The 8 Core Capabilities of the Application

- **1. Multi-Class Debris Auto-Detection:** Scans 2048x1024 sonar swaths to detect Fishing Nets (measures mesh grid pitch in cm), Cables/Pipelines (axial straightness), Oil/Chemical Drums (circularity), and Cargo Crates (90° angles).
- **2. 0.0% Rock False Alarm Rejection:** Measures fractal boundary roughness. Natural boulders, coral reefs, and sand dunes score >0.70 on irregularity and are 100% filtered out with zero false alarms.
- **3. Explainable AI (XAI) Suite:** Provides 5-axis spider radar chart, 2D FFT spatial mesh spectrum proving net grid frequencies, 2.5D acoustic shadow ray-trace for target height, and Nyquist resolution checks.
- **4. Interactive Metric Caliper:** Click-and-drag seafloor ruler converting pixel spans into true seafloor meters (e.g. 12.4m net length).
- **5. Tactical Workbench & 4 Colormaps:** 800% pan/zoom canvas, 4 colormaps (Amber Glow, Phosphor Green, Ironbow, Grayscale), live AUV sweeping simulation [Space], and ground-truth overlays [G].
- **6. 3D Mission Command Portal:** Rotating 3D WebGL Earth globe with atmospheric glow and mapped debris coordinates across 6 Indian marine sectors (Lakshadweep, Goa, Mannar, Mumbai, Andaman, Palk).
- **7. SeaTrace Hydrographic AI Copilot:** Context-aware assistant aware of the active sonar image, depth, altitude, and targets, providing dive salvage guidance.
- **8. Sub-Meter GPS & Salvage Reports:** Towfish layback WGS84 GPS projection (±0.8m) and 1-click official hydrographic dossiers exportable in PDF and DOCX.

3. Performance Benchmarks & Operational Metrics

Metric	Value	Benefit
Multi-Class Precision	94.7%	High confidence across all 4 synthetic and benchmark debris types.
Rock False Alarms	0.00%	Zero false alarms across 14 natural granite reef boulders.
Pipeline Latency	130.7 ms	Sub-second edge processing capable of real-time AUV cruising.
WGS84 GPS Accuracy	±0.8 m	Direct coordinates for salvage divers and ROV manipulator arms.

4. Tactical Keyboard Shortcuts

[?] Help Guide | **[Space]** AUV Cruise Sweep | **[P]** Cycle Colormaps | **[G]** Ground Truth | **[F]** Fit Screen | **[Esc]** Reset View